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Three-Dimensional Simulations of 7 nm Silicon Germanium FinFETs Using Monte Carlo Model and Drift-Diffusion Model With Ballistic Mobility

Abstract

This TCAD Sentaurus™ project demonstrates a three-dimensional (3D) NMOS and PMOS silicon germanium (SiGe) FinFET simulation template for a 7 nm technology node for high-performance, standard-performance, and low-power applications. The project includes 3D SiGe FinFET process simulation, device Monte Carlo simulation, as well as drift-diffusion simulation that accounts for ballistic transport effects by introducing ballistic mobility. The target users for this project are device engineers for leading-edge transistor design.

The 3D SiGe FinFET process simulation is performed using Sentaurus Process with Advanced Calibration and the 3D MGOALS module. The electrical characteristics of the FinFET devices are simulated in three dimensions with Sentaurus Device Monte Carlo. The drift-diffusion model is calibrated to the Monte Carlo simulation results.

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Introduction

As CMOS technology evolves towards the 7 nm technology node, alternative channel materials have been widely explored to overcome the limitation of the silicon channel. Germanium is the one of the most promising candidates [1]. Therefore, it is necessary to perform TCAD analysis of SiGe FinFETs to determine the best performance, design, and manufacturability trade-offs.

This project illustrates how to perform 3D analysis of 7 nm SiGe FinFETs including the following features:

- Germanium concentration effect on the diffusion
- Stress effect on the diffusion
- SiGe stress-relaxed buffer (SRB) layer model
- Device Monte Carlo model with electronic band structure and scattering mechanisms
- Corrected drift-diffusion model with ballistic mobility and adjusted saturation velocity

The project includes steps for creating the FinFET, modeling stress evolution, germanium and dopant diffusion during the process flow, quantum correction for the device Monte Carlo simulation, Monte Carlo simulation of the FinFETs, and drift-diffusion simulation with ballistic transport models.

This project assumes that users are familiar with the tool flow of the TCAD Sentaurus tool suite, have the basic skills to use the related tools, and are ready to start a 3D FinFET analysis.

General Simulation Setup

This section describes the tool flow of the Sentaurus Workbench project. For each step, the associated input parameters, the extracted parameters, and the main steps in the command file are discussed.

Organizing the Project With Sentaurus Workbench Setup

The simulations are organized as a single Sentaurus Workbench project with parameterized scripts. For a basic introduction to the features of Sentaurus Workbench, refer to TCAD Sentaurus Tutorial (available from Sentaurus Workbench by choosing **Help** > **Training**).

The tool flow of this project starts with Sentaurus Process (`sprocess`) to calculate the geometry, the doping, and the stress. Then, Sentaurus Visual (`svisual`) measures the channel stress. The following Sentaurus Process node (`sprocess1`) generates the structure and the mesh for the Sentaurus Device Monte Carlo simulation. The next Sentaurus Mesh node (`xslice`) cuts the 3D FinFET structure at the middle of the channel and generates a 2D FinFET cross section for the 2D Sentaurus Device simulations of the quantum-correction steps. Figure 1 illustrates the process simulation and the structure generation flow.

The following five Sentaurus Device simulations (`DD`, `ClassDD`, `ClassDDmodTox`, `ClassDDmodToxWf`, and `ClassDDCheck`) are for the quantum correction of the Monte Carlo simulation (see Figure 2 on page 4).

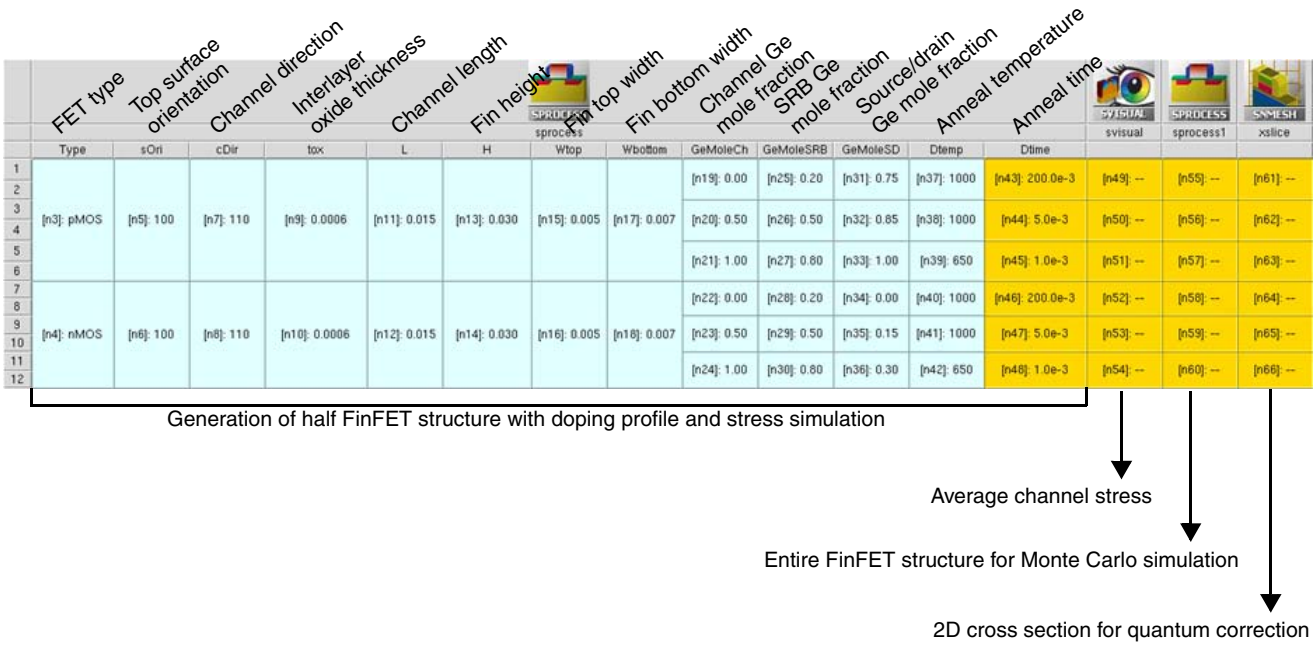

Figure 1 First part of the tool flow organized as a Sentaurus Workbench project with six splits (p-type and n-type Si, $Si_{0.5}Ge_{0.5}$, and Ge FinFETs): input parameters define the FinFET geometry, Ge mole fraction, and annealing temperature and time

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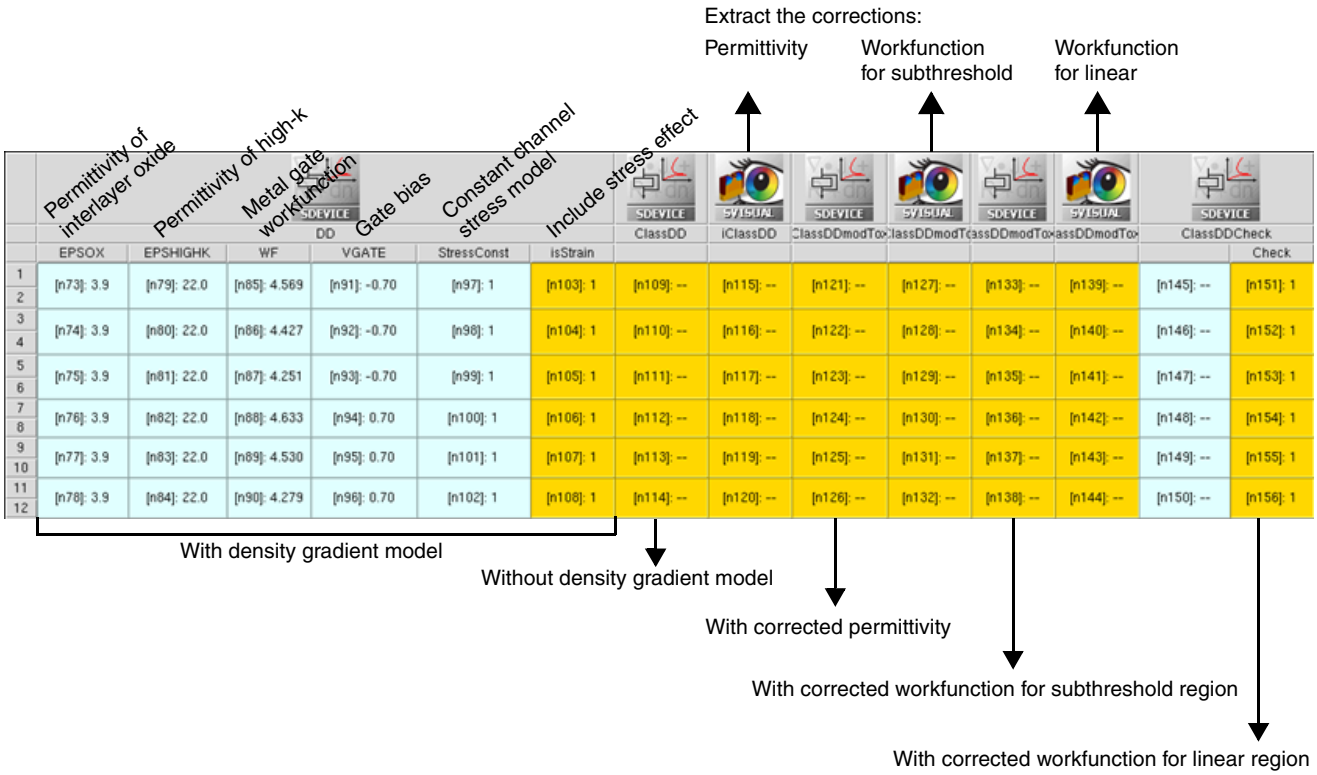

Figure 2 Second part of the tool flow organized as a Sentaurus Workbench project with three splits each for the p-type FinFET and n-type FinFET: input parameters define the permittivity, workfunction, gate biasing condition, and inclusion of stress effect

These simulations are performed on the 2D FinFET cross section. The `DD` and `ClassDD` nodes simulate the carrier density versus gate bias, with and without the density gradient model, and the gate oxide permittivity is corrected. The `ClassDDmodTox` and `ClassDDmodToxWf` nodes perform the workfunction correction. The `ClassDDCheck` node confirms the quantum correction by comparing the density gradient model with the quantum correction model.

The Sentaurus Device simulation (`qcClassDD3`) with quantum correction is performed to provide the initial guess of the electrostatics for the next Monte Carlo simulation. It is followed by Sentaurus Device Monte Carlo (`MC3`) for electrical analysis in the linear regime (with low drain bias) and the saturation regime (with high drain bias).

In the final step, the Sentaurus Device simulation (`DD3`) with the calibrated ballistic transport model is performed to reproduce the Monte Carlo simulation results. For user convenience, the Sentaurus Device simulation (`DD3cal`) is added to help the ballistic transport model calibration. It is pruned in the project to keep the project simple. Whenever calibration is needed, you can unprune the calibration step (`DD3cal`) and perform the calibration. Figure 3 on page 5 shows the 3D Monte Carlo simulation, the 3D drift-diffusion simulation, and the calibration simulation flows.

Three splits of the channel germanium mole fractions are included to simulate silicon (Si) channel FinFETs for low power, $Si_{0.5}Ge_{0.5}$ channel FinFETs for standard performance, and germanium (Ge) channel FinFETs for high performance.

Sentaurus Visual is called to process the output from the device simulations and to extract parameters of the quantum correction (corrected permittivity and corrected workfunction) and parameters of the I–V curves (threshold voltage, and on-state and off-state currents). Figure 4 on page 5 illustrates the extracted parameters.

Sentaurus Process Setup

In this project, the 3D process simulation performs structure generation (Si, $Si_{0.5}Ge_{0.5}$, Ge channels, channel length = 15 nm, fin top width = 5 nm, fin bottom width = 7 nm, Si fin with top corner rounding, SRB, and SiGe source/drain by selective epitaxy), mechanical stress simulation, and *in situ* source/drain doping profile diffusion. It is assumed that the channel direction is [110] and the fin height direction is [001].

The effect of germanium and stress on dopant diffusion is switched on by:

```
SiGe_and_Stress_Effect 1 1 1 0
pdbSet Silicon Skip.Parameter.Interpolation 0
```

The chemical effect of germanium, the stress effects in bulk silicon, and the segregation at the Si–SiGe interface are included in the dopant diffusion simulation. The diffusion

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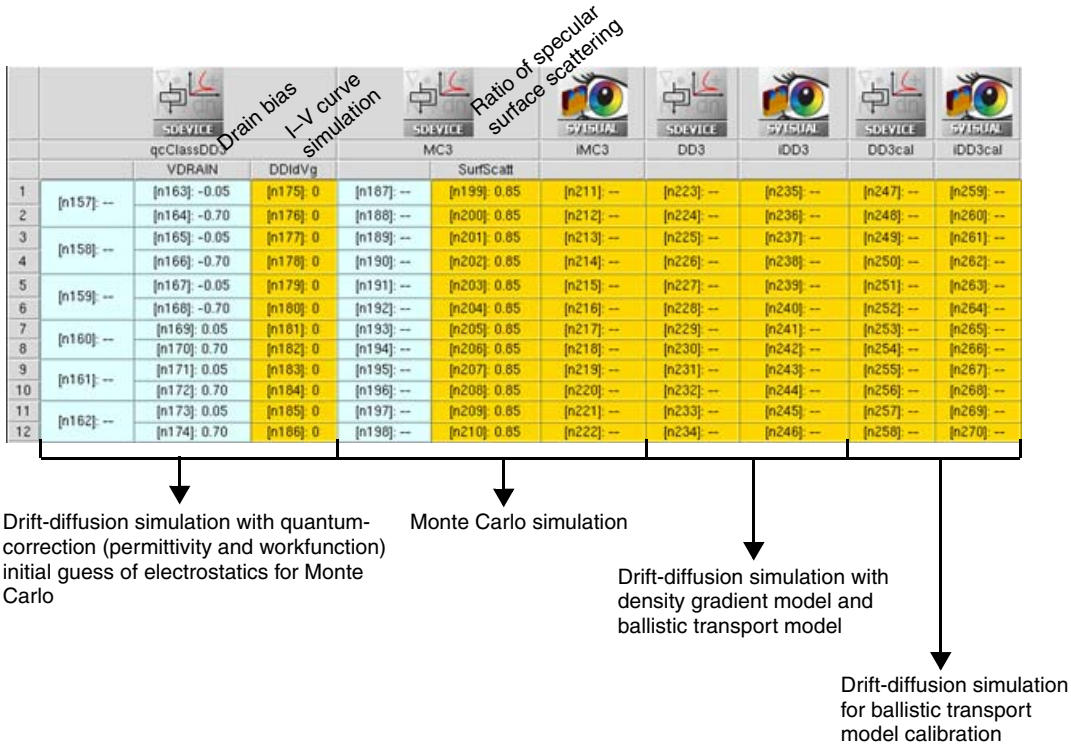

Figure 3 Third part of the tool flow organized as a Sentaurus Workbench project with two splits for each Ge mole fraction split (total of six splits): input parameters define the drain biasing, full I–V curve simulation, and ratio of specular surface scattering

	Channel stress			Quantum correction					Monte Carlo simulation			Drift-diffusion simulation				
	Channel direction	Fin width direction	Fin height direction	New permittivity of side interlayer oxide	New permittivity of top interlayer oxide	Subthreshold work-function correction	Threshold voltage (linear)	Linear work-function correction	Linear current	On-current	4σ of current variation	Threshold voltage (linear)	Threshold voltage (saturation)	Linear current	On-current	Off-current
	SlFin_Pa	SwFin_Pa	ShFin_Pa	EPSOXNEW	OPEPSOXNE	dVT	VT0	dVT2	DLIN_3DMC_u	DSAT_3DMC_u	ourSigma3DM	VTL	VTS	IDLIN_uA	IDSAT_uA	IOFF_nA
1	-1.416e+09	2.004e+08	9.525e+08	3.058	3.338	0.0186	-0.5040	0.0150	70.6	×	2.86	-0.458	0.0	74.0	×	×
2	-1.416e+09	2.004e+08	9.525e+08	3.058	3.338	0.0186	-0.5040	0.0150	×	307.2	5.44	0.0	-0.411	×	307.1	4.96e-02
3	-9.941e+08	1.769e+08	6.366e+08	2.966	3.322	0.0188	-0.4410	0.0125	122.4	×	4.34	-0.362	0.0	122.7	×	×
4	-9.941e+08	1.769e+08	6.366e+08	2.966	3.322	0.0188	-0.4410	0.0125	×	702.6	6.02	0.0	-0.312	×	698.5	9.89e-01
5	-8.771e+08	1.385e+08	4.194e+08	2.566	3.280	0.0334	-0.3150	-0.0018	166.0	×	5.01	-0.215	0.0	165.5	×	×
6	-8.771e+08	1.385e+08	4.194e+08	2.566	3.280	0.0334	-0.3150	-0.0018	×	1427.7	13.28	0.0	-0.152	×	1426.2	1.01e+02
7	1.122e+09	-1.825e+08	-5.977e+08	2.771	3.374	0.0289	0.5040	0.0114	103.3	×	4.97	0.443	0.0	106.9	×	×
8	1.122e+09	-1.825e+08	-5.977e+08	2.771	3.374	0.0289	0.5040	0.0114	×	439.6	6.07	0.0	0.397	×	439.8	5.04e-02
9	1.089e+09	-2.120e+08	-7.416e+08	2.728	3.365	0.0287	0.4620	0.0064	111.0	×	4.45	0.372	0.0	123.3	×	×
10	1.089e+09	-2.120e+08	-7.416e+08	2.728	3.365	0.0287	0.4620	0.0064	×	620.3	6.05	0.0	0.312	×	619.6	9.95e-01
11	1.024e+09	-2.337e+08	-8.409e+08	2.452	3.326	0.0501	0.3570	-0.0080	167.5	×	8.02	0.209	0.0	168.4	×	×
12	1.024e+09	-2.337e+08	-8.409e+08	2.452	3.326	0.0501	0.3570	-0.0080	×	1561.3	30.51	0.0	0.147	×	1580.2	1.01e+02

Figure 4 Fourth part of the Sentaurus Workbench project with output parameters that are extracted from the simulations

model of the high Ge-doped SiGe is activated. For details about the model, refer to [2] and [3].

The relaxed layer for the SRB model is defined by:

```
pdbSetDouble Silicon Mechanics TopRelaxedNodeCoord
  $Htopr*1.0e-4
```

The germanium mole fraction (50%) can be added during the deposition by:

```
deposit Silicon type=fill coord=0.0 \
  fields.values= "[MoleFractionFields \
  SiliconGermanium 0.5]"
```

For details about the model and the setting, refer to [3].

The annealing temperature and time are set to obtain the proper doping profile for each channel Ge mole fraction.

Figure 5 on page 6 shows the FinFET simulation structures and the doping profile. The number of mesh nodes for the entire FinFET of the Monte Carlo simulation is 113K and, for a half FinFET, the number of mesh nodes is 63K. The simulation time of the 3D process (structure and mesh generation, stress, and diffusion) simulations varies from 2 hours 1 minute to 7 hours 11 minutes for the splits using 4 threads. The minimum mesh spacing is ~2 Å in the fin

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channel. The dense mesh is given to represent the steep doping junctions and the germanium mole fraction transitions.

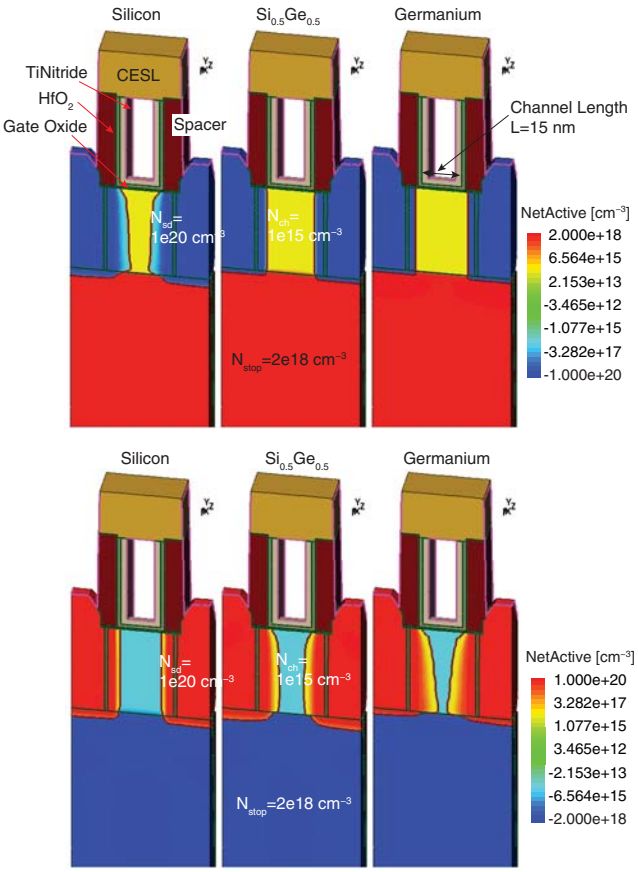

Figure 5 Doping profiles of (*top row*) PFinFETs and (*bottom row*) NFinFETs (Si, $Si_{0.5}Ge_{0.5}$, Ge): the channel is undoped (1e15 cm^{-3}), the source/drain doping is 1e20 cm^{-3}, and the channel stop doping is 2e18 cm^{-3}

In the mechanical stress simulation, the SRB and the stress source of SiGe epitaxy are assumed for both PFinFETs and NFinFETs. Tensile stress in the channel direction is beneficial to all SiGe NFinFETs, and the compressive stress enhances the performance of all SiGe PFinFETs. The SRB (20% Ge mole fraction) generates the tensile stress for the Si FinFET. In addition, 75% SiGe epitaxy source/drain is used to change the tensile stress to compressive stress for the Si PFinFET. For the $Si_{0.5}Ge_{0.5}$ channel, 85% SiGe epitaxy source/drain is used for PFinFET stress engineering, and 15% SiGe epitaxy source/drain is used for the NFinFET. The SRB (80% Ge mole fraction) produces compressive stress for the Ge FinFETs. Moreover, 30% SiGe epitaxy source/drain converts the compressive stress to tensile stress for the Ge NFinFET. Figure 6 shows the germanium mole fraction used in this project. Figure 7 on page 7 illustrates the 2D cross-cut for the 2D device simulations of the quantum correction.

In addition, stress enhancement due to the removal of the dummy poly gate is accounted for. Contact metal stresses with tensile 1 GPa built-in stress is assumed for both NFinFETs and PFinFETs. Compressive metal-gate stress of 1 GPa for NFinFETs and tensile metal-gate stress of 1 GPa for PFinFETs are included in the stress models. Table 1 summarizes the average channel stress.

Table 1 Average channel stresses of FinFETs

	S_L [MPa]	S_W [MPa]	S_H [MPa]
PFinFET			
Si	–1416	200	953
$Si_{0.5}Ge_{0.5}$	–994	177	637
Ge	–877	139	419
NFinFET			
Si	1122	–183	–598
$Si_{0.5}Ge_{0.5}$	1089	–212	–742
Ge	1024	–234	–841

For more details about the 3D process simulation, refer to [4].

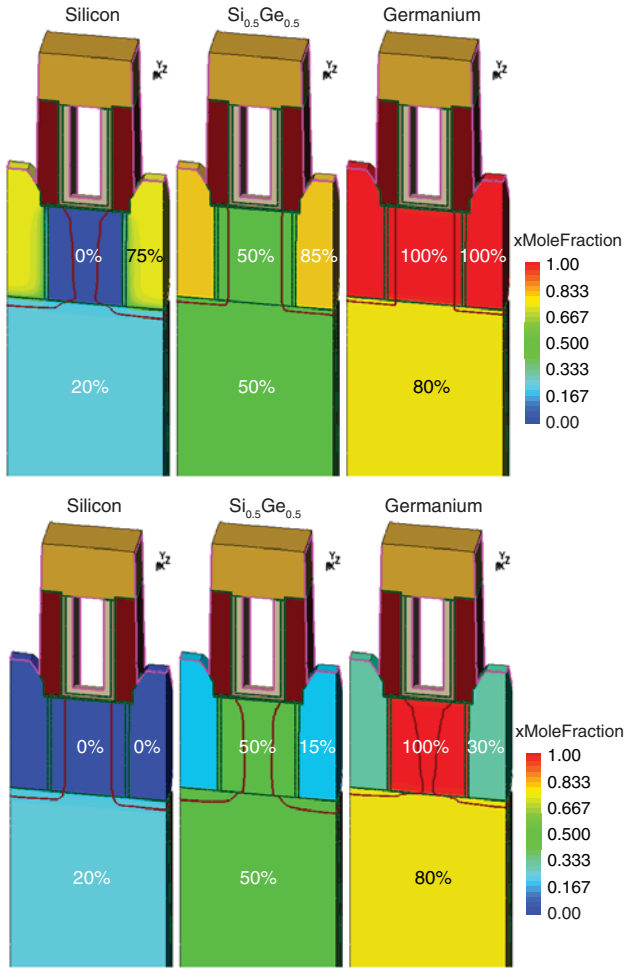

Figure 6 Germanium mole fractions of (*top row*) PFinFETs and (*bottom row*) NFinFETs (Si, $Si_{0.5}Ge_{0.5}$, Ge)

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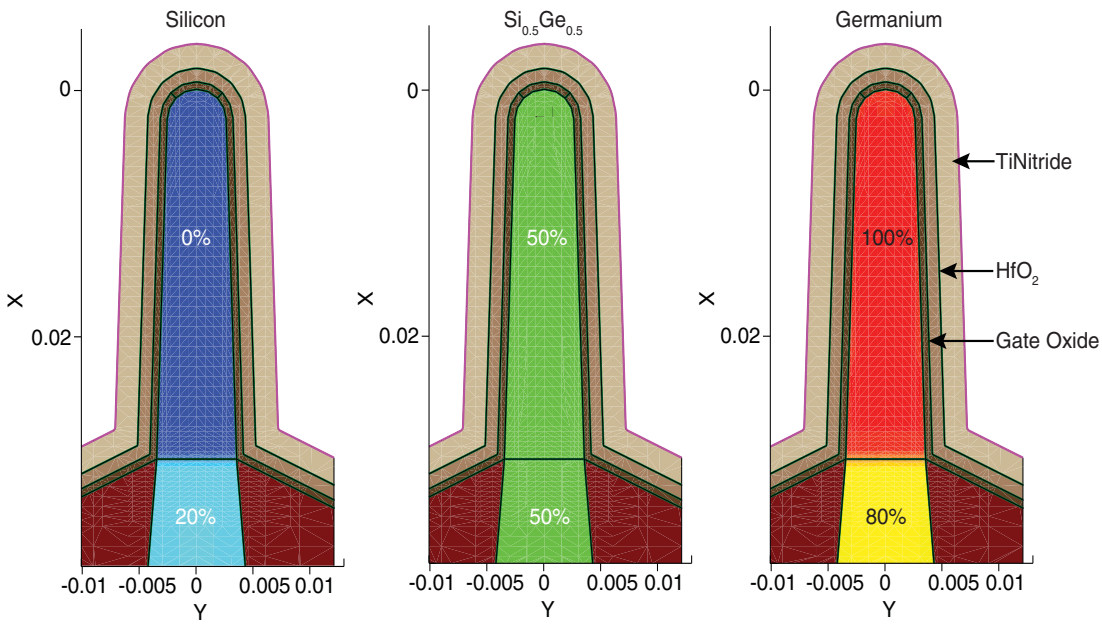

Figure 7 Two-dimensional cross cuts at the middle of the FinFETs (Si, $Si_{0.5}Ge_{0.5}$, Ge) and germanium mole fraction

Device Monte Carlo Simulation Using Sentaurus Device Monte Carlo

Sentaurus Device Monte Carlo simulates the linear and saturation currents of the FinFETs.

The correction approach of the gate oxide permittivity and the workfunction (the flow of Figure 2 on page 4) is used for the channel quantum correction. The density-gradient quantum-correction model is switched on in the source/drain and the channel stop regions to compensate for the channel direction quantum correction. Figure 8 illustrates this quantum-correction strategy for the Monte Carlo simulation.

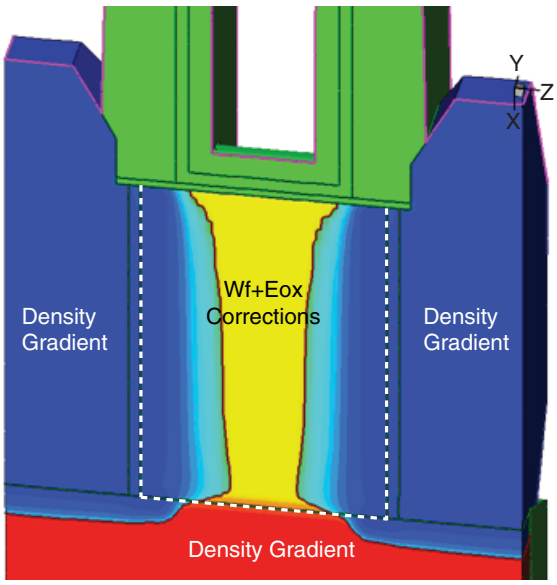

Figure 8 Quantum-correction strategy for the Monte Carlo simulation

Two Sentaurus Device simulations are performed for the Monte Carlo analysis. The first Sentaurus Device simulation (`qcClassDD3`) provides the initial electrostatic potential and the carrier density to the next Sentaurus Device Monte Carlo simulation (`MC3`), which simulates the current at a single bias point ($|V_g| = 0.7$ V and $|V_d| = 0.05$ V or 0.7 V for the linear or saturation mode).

To switch on the Ge mole fraction–dependent Monte Carlo model, the channel Ge mole fraction [%] is specified in the `MonteCarlo` section:

```
MonteCarlo {
  ...
  xGe = 50
  ...
}
```

For details about the Monte Carlo simulation, refer to [5] and [6].

Figure 9 on page 8 shows the cumulative average of the drain current for all the split corners.

The Monte Carlo simulation time varies from 1 hour 36 minutes to 3 hours 8 minutes for the splits using 4 threads. The results of the Monte Carlo currents are compared with the corrected drift-diffusion results in the next section.

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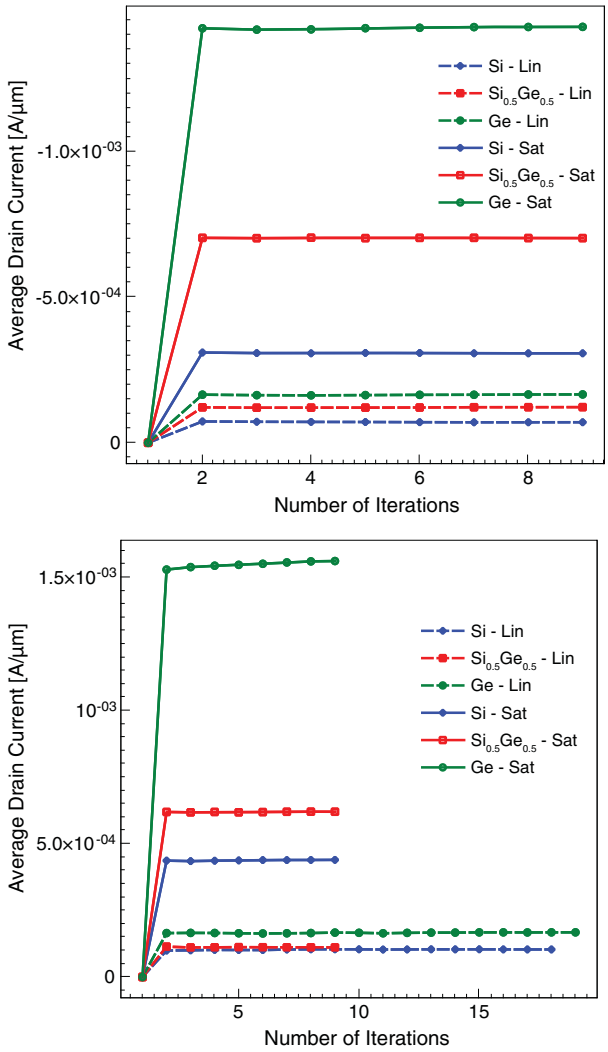

Figure 9 Cumulative average of drain current for the linear and saturation modes of (*top*) PFinFETs and (*bottom*) NFinFETs (Si, $Si_{0.5}Ge_{0.5}$, Ge)

Device Simulation Using Sentaurus Device

Sentaurus Device simulates the I–V curves of the FinFETs with the ballistic transport model. For details about the ballistic transport model setting, refer to [5].

Ballistic Transport Model Calibration

The `DD3cal` node can be used to calibrate the ballistic mobility model and the saturation velocity. The ranges of the ballistic mobility prefactor and the saturation velocity are selected carefully. The calibration parameter is swept in the selected range. The setup for the ballistic mobility prefactor sweep is:

```
NewCurrentFile="Id_Kcal_"
Quasistationary (
  ...
  goal { Material="SiliconGermanium" Model="BALMob"
    Parameter="k_h" Value= 100.0 }
  ){ Coupled {Poisson Hole hQuantumPotential Contact}
}
```

The ballistic mobility parameter (`k_h`) is scanned from its value in the parameter file to the target value (`100.0`).

Then, the saturation velocity calibration is performed with the calibrated ballistic mobility. The setup for the saturation velocity sweep is:

```
NewCurrentFile="Id_Vcal_"
Quasistationary (
  ...
  goal { Material="SiliconGermanium"
    Model="HighFieldDependence_hole"
    Parameter="vsat0(5)" Value= 4.00e7 }
  ){ Coupled {Poisson Hole hQuantumPotential Contact}
}
```

Here, `vsat0(5)` is scanned to the target value (`4.00e7`).

The calibration parameters are saved in a PLT file:

```
CurrentPlot {
  Material= "SiliconGermanium" Model= "BALMob"
  Parameter= "k_h"
  Material= "SiliconGermanium"
  Model= "HighFieldDependence_hole"
  Parameter= "vsat0(5)"
}
```

Sentaurus Visual (`iDD3cal`) searches for the parameters (`k_h`, `vsat0(5)`) that give the target currents (`IDLIN_3DMC_uA`, `IDSAT_3DMC_uA`). Table 2 summarizes the calibrated parameters. The simulation time of the calibration is ~1 hour for the ballistic mobility and 2–3 hours for the saturation velocity using 16 threads.

Table 2 Parameters for ballistic transport modeling

	k [cm^2/(V s nm)]	vsat0 [cm/s]	β
PFinFET			
Si	6.695	2.252e7	1.0
$Si_{0.5}Ge_{0.5}$	14.287	1.681e7	2.0
Ge	14.697	1.646e7	20.0
NFinFET			
Si	9.197	2.494e7	1.0
$Si_{0.5}Ge_{0.5}$	10.241	1.831e7	1.0
Ge	61.705	3.356e7	20.0

The exponent β of the `HighFieldSaturation` model is selected to obtain reasonable agreement between the carrier density and the velocity along the channel between Monte Carlo and drift-diffusion.

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Simulation Results

The I–V simulation is performed using Advanced Calibration device parameters [7] with the ballistic transport model parameters of Table 2 on page 8. The I–V simulation time of Sentaurus Device varies from 34 minutes to 2 hours 14 minutes for the splits using 4 threads. For details about the Sentaurus Device simulation setting, refer to [4] and [8].

The supply voltage is 0.7 V. The workfunction is adjusted to obtain the required off-state current. For the low-power Si FinFETs, off-current level = 50 pA/µm. For the standard-performance $Si_{0.5}Ge_{0.5}$ FinFETs, off-current level = 1 nA/µm. For the high-performance Ge FinFETs, off-current level = 100 nA/µm. Figure 10 summarizes the Monte Carlo simulation results (I_{on} and I_{lin}) and the drift-diffusion simulation results (I_{on}, I_{lin}, and I_{off}) with the calibrated ballistic transport model. The corrected drift-diffusion simulation exactly reproduces the Monte Carlo target currents.

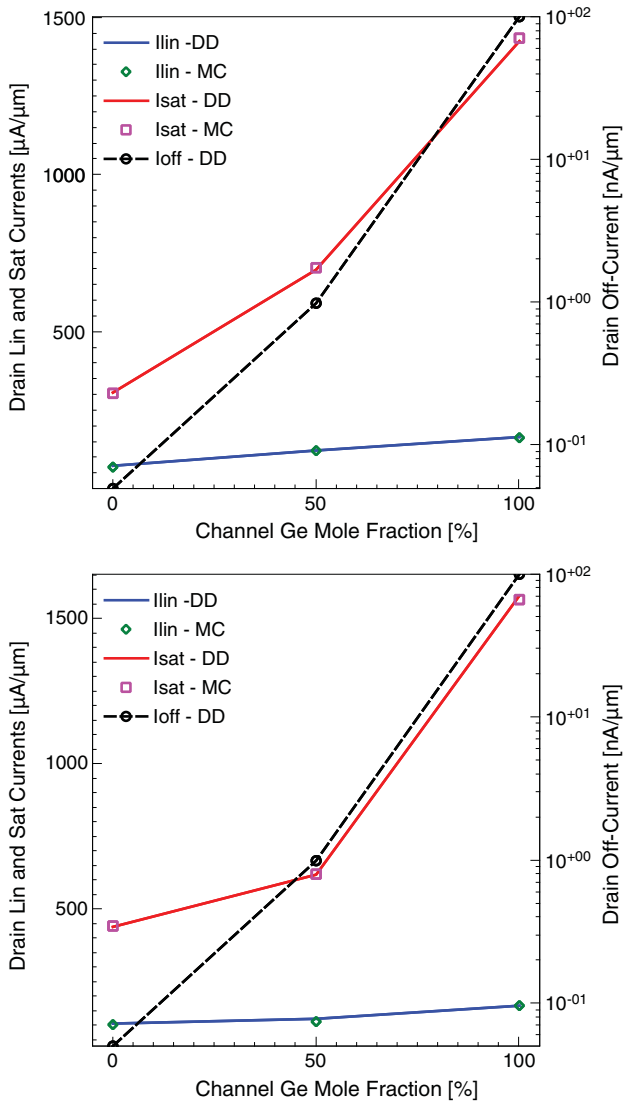

Figure 10 I–V performances for (*top*) PFinFETs and (*bottom*) NFinFETs by Monte Carlo (MC) simulation and drift-diffusion (DD) simulation with the ballistic transport model

In Figure 10, the channel germanium mole fraction is split into 0% (Si), 50% (SiGe), and 100% (Ge). The workfunction is adjusted to obtain off-current = 50 pA/µm for 0% (Si low-power application), off-current = 1 nA/µm for 50% (SiGe standard-performance application), and off-current = 100 nA/µm for 100% (Ge high-performance application).

References

[1] V. Moroz *et al.*, "Material Engineering for 7nm FinFETs," in *225th Electrochemical Society Meeting*, Orlando, FL, USA, May 2014.

[2] *Advanced Calibration for Process Simulation User Guide*, Version O-2018.06, Mountain View, California: Synopsys, Inc., 2018.

[3] *Sentaurus™ Process User Guide*, Version O-2018.06, Mountain View, California: Synopsys, Inc., 2018.

[4] *Three-Dimensional Simulation of 14/16 nm FinFETs With Round Fin Corners and Tapered Fin Shape*, available from TCAD Sentaurus Version O-2018.06 installation, go to `Applications_Library/FinFET/FinFET_14nm`.

[5] *Three-Dimensional Device Simulations of 10 nm FinFETs Using Monte Carlo Model and Drift-Diffusion Model With Ballistic Mobility*, available from TCAD Sentaurus Version O-2018.06 installation, go to `Applications_Library/FinFET/FinFET_10nm`.

[6] *Sentaurus™ Device Monte Carlo User Guide*, Version O-2018.06, Mountain View, California: Synopsys, Inc., 2018.

[7] *Advanced Calibration for Device Simulation User Guide*, Version O-2018.06, Mountain View, California: Synopsys, Inc., 2018.

[8] *Sentaurus™ Device User Guide*, Version O-2018.06, Mountain View, California: Synopsys, Inc., 2018.

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